Countable Additivity is Not First-Order:
A Companion Note to "Probability from Observation: Collective Exhaustion and the Observable Extension Theorem"

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Abstract

Proposition 3.9 of the companion paper shows that collective exhaustion (CE) is not derivable from any first-order structural condition on a query system. We show here that this obstruction is not specific to query systems. In the first-order language of Boolean algebras equipped with a normalized finitely additive charge, no first-order theory characterizes those models whose charge extends to a σ-additive measure on the generated σ-algebra. Equivalently, countable additivity is not first-order axiomatizable in this language. The proof uses the same ultraproduct mechanism as Proposition 3.9, now stated in full generality.

1 The language

Let $\mathcal{L}_{\mathrm{BA},\mu}$ be the one-sorted first-order language with:

- Boolean operations $\wedge, \vee, \neg, 0, 1$;
- for each rational $q \in [0,1] \cap \mathbb{Q}$, unary relation symbols $M_{\leq q}(x)$ and $M_{\geq q}(x)$, intended to mean $\mu(x) \leq q$ and $\mu(x) \geq q$ respectively.

A structure for $\mathcal{L}_{\mathrm{BA},\mu}$ is a Boolean algebra B together with a $[0,1]$-valued finitely additive probability μ, encoded through the truth of the predicates $M_{\leq q}$ and $M_{\geq q}$. Using rational comparison predicates rather than a function symbol into $[0,1]$ avoids introducing a second numeric sort.

Definition 1.1 (First-order theory of finitely additive probabilities). Let T_{fa} be the $\mathcal{L}_{\mathrm{BA},\mu}$-theory axiomatizing:

(i) B is a Boolean algebra;

(ii) normalization: $\mu(0) = 0$ and $\mu(1) = 1$;

(iii) monotonicity: $x \leq y \Rightarrow \mu(x) \leq \mu(y)$;

(iv) finite additivity: for all rationals r, s with $r + s \leq 1$,

$$x \wedge y = 0 \wedge M_{\leq r}(x) \wedge M_{\leq s}(y) \;\Rightarrow\; M_{\leq r+s}(x \vee y),$$

and similarly for lower bounds.

The class of Boolean algebras with normalized finitely additive probability is first-order axiomatizable in $\mathcal{L}_{\mathrm{BA},\mu}$. Countable additivity, by contrast, requires quantification over an arbitrary countable sequence $(a_n)_{n\in\mathbb{N}}$ — not available in first-order logic — and so is not a sentence of $\mathcal{L}_{\mathrm{BA},\mu}$.

2 The metatheorem

Proposition 2.1 (σ-additive extension is not first-order)**.** *There is no first-order $\mathcal{L}_{\mathrm{BA},\mu}$-theory $T \supseteq T_{\mathrm{fa}}$ such that, for every $(B,\mu) \models T_{\mathrm{fa}}$,*

$$(B,\mu) \models T \quad \Longleftrightarrow \quad \mu \text{ extends to a } \sigma\text{-additive measure on } \sigma(B).$$

Equivalently, countable additivity is not first-order axiomatizable among finitely additive probability Boolean algebras.

Proof. Let E be the Boolean algebra of finite-cofinite subsets of $\mathbb{Q}$, fix an enumeration $q : \mathbb{N} \to \mathbb{Q}$, and let δ_n be the Dirac charge at $q(n)$. Each (E, δ_n) is a model of T_{fa} in which δ_n is σ-additive. Suppose for contradiction that a theory T as described exists; then each $(E, \delta_n) \models T$.

Let $\mathcal{U}$ be a nonprincipal ultrafilter on $\mathbb{N}$ extending the cofinite filter. By Łoś's theorem [1], the ultraproduct $\prod_{\mathcal{U}}(E, \delta_n)$ also models T. The charge induced on the diagonal copy of E is

$$\ell(A) = \lim_{\mathcal{U}} \delta_n(A) = \lim_{\mathcal{U}} \mathbf{1}_{q(n)\in A}.$$

If A is finite, $q(n) \in A$ for only finitely many n, so $\ell(A) = 0$. If A is cofinite, $q(n) \in A$ for cofinitely many n, so $\ell(A) = 1$ since $\mathcal{U}$ extends the cofinite filter. Thus ℓ is the finite-cofinite charge.

The sequence $A_k = \mathbb{Q} \setminus \{q(0), \dots, q(k)\}$ satisfies $A_0 \supseteq A_1 \supseteq \cdots$ and $\bigcap_k A_k = \varnothing$, yet $\ell(A_k) = 1$ for all k. Hence the charge induced on the diagonal copy of E fails σ-additivity and does not extend to a σ-additive measure on $\sigma(E)$. Consequently the ultraproduct cannot satisfy any theory T characterizing σ-additive extension, contradicting Łoś's theorem applied to the assumption that each $(E, \delta_n) \models T$. $\square$

Remark 2.2. For precision: the ultraproduct $\prod_{\mathcal{U}}(E, \delta_n)$ is an $\mathcal{L}_{\mathrm{BA},\mu}$-structure in its own right; ℓ is the charge induced on the diagonal copy of E inside it. Identifying the ultraproduct with (E, ℓ) is harmless for the contradiction, since the relevant failure of σ-additivity is witnessed on the diagonal copy.

3 Corollary in the language of collective exhaustion

In the companion paper [2], collective exhaustion (CE) is the exact condition under which a compatible finitely additive family extends to a σ-additive measure on the observable σ-algebra. Proposition 2.1 therefore implies:

Corollary 3.1 (CE is not first-order in the general language)**.** *In the first-order language $\mathcal{L}_{\mathrm{BA},\mu}$ of Boolean algebras with normalized finitely additive charge, no first-order $\mathcal{L}_{\mathrm{BA},\mu}$-condition characterizes, or in general forces, the extension property characterized in the companion paper by collective exhaustion.*

The companion paper establishes this at the level of query systems (Proposition 3.9): no first-order structural condition in the query-system language implies CE. Corollary 3.1 shows the obstruction is not an artifact of the query-system formalism. It already appears in the more primitive setting of Boolean algebras with finitely additive charges, where countable additivity is not first-order axiomatizable.

The failure of CE to be structurally derivable therefore reflects a genuine expressive limitation of finitary first-order description: such descriptions can encode finite coherence, but not the countable-exhaustion behavior required for probability.

The philosophical disagreements about the foundations of probability are not terminological. They are disagreements about which extra structure licenses the passage from finite coherence to probability — and the present result shows that such extra structure cannot be avoided. Countable additivity cannot be derived from finite coherence alone; it is a genuine additional condition. This helps explain why foundational disputes about probability persist: they turn on what must be added, and why, to obtain σ-additive probability from finite coherence.

4 The representation question

Proposition 2.1 shows that *one* purely finitely additive charge arises as an ultraproduct of σ-additive ones — that single witness suffices for the non-axiomatizability conclusion. A stronger question is:

> *Which purely finitely additive charges arise as ultraproducts or ultralimits of σ-additive probabilities?*

This is a representation question, and it is strictly harder than non-axiomatizability. It may hold only after enlarging the notion of approximation, for example by allowing nets or ultralimits more general than literal ultraproducts. The Yosida–Hewitt decomposition [3] identifies the purely finitely additive part ℓ_p of any finitely additive charge; whether ℓ_p can always be represented as an ultralimit is not settled here and is left as an open question.

References

[1] Jerzy Łoś. Quelques remarques, théorèmes et problèmes sur les classes définissables d'algèbres. In *Mathematical Interpretation of Formal Systems*, pages 98–113. North-Holland, 1955.

[2] Patrick S. Mahon. Probability from observation: Carathéodory and Stone routes to the observable extension theorem. Preprint, 2026.

[3] Kôsaku Yosida and Edwin Hewitt. Finitely additive measures. *Transactions of the American Mathematical Society*, 72(1):46–66, 1952.